GROCERY BILLING SYSTEM

1. Title

Project Title: Grocery Billing System
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2. Abstract

The project **Grocery Billing System** is a simple console-based application developed in the C programming language. It helps in automating the process of generating bills for grocery products. The program allows the user to input product details such as name, price, and quantity, then calculates the total and applies discounts if applicable. This project demonstrates fundamental programming concepts like loops, conditional statements, and arithmetic operations.

3. Introduction

The Grocery Billing System aims to simplify the manual process of billing in small retail or grocery shops. It calculates the total cost of products purchased and applies discounts when the total exceeds a specified amount. This project is useful for beginners to understand structured programming and can be used as an educational tool to learn about real-world applications of C programming.

4. Objectives

- To understand structured programming concepts.
- To use loops and conditional statements effectively.
- To apply arithmetic operations for billing calculations.
- To improve logical thinking and problem-solving skills.

5. Tools & Technology

- **Programming Language:** C
- **Compiler:** GCC / Turbo C / Code::Blocks

- **Platform:** Windows / Linux

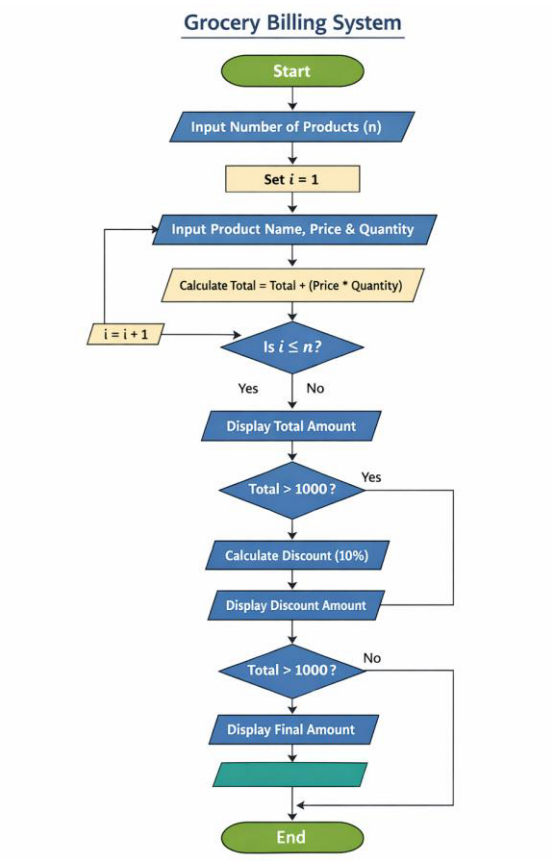
6. System Requirements

- **Hardware:** Basic computer with minimum 4GB RAM
- **Software:** C Compiler and any IDE/Text Editor

7. Methodology / Algorithm

1. Start program
2. Display title "GROCERY BILLING SYSTEM"
3. Read the number of products
4. For each product:
    - Read product name, price, and quantity
    - Calculate total cost
5. If total > 1000, apply 10% discount
6. Display total, discount, and final amount
7. Display thank you message
8. End program

8. Flowchart



9. Program Code

```
#include <stdio.h>
#include <string.h>

int main() {
    char product[50];
    int quantity;
    float price, total = 0, discount = 0, finalAmount;
    int n, i;

    printf("===== GROCERY BILLING SYSTEM =====\n");

    printf("Enter number of products: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        printf("\nProduct %d\n", i);
        printf("Enter product name: ");
        scanf("%s", product);
        printf("Enter price: ");
        scanf("%f", &price);
        printf("Enter quantity: ");
        scanf("%d", &quantity);

        total = total + (price * quantity);
    }

    printf("\n----------------------------------");
    printf("\nTotal Amount = %.2f", total);

    // Discount condition
    if (total > 1000) {
        discount = total * 0.10;  // 10% discount
        printf("\nDiscount (10%%) = %.2f", discount);
    }

    finalAmount = total - discount;

    printf("\nFinal Amount = %.2f", finalAmount);
    printf("\n----------------------------------");
    printf("\nThank You! Visit Again\n");

    return 0;
}
```

10. Sample Input / Output

Output:

```
===== GROCERY BILLING SYSTEM =====
Enter number of products: 2

Product 1
Enter product name: Shampoo
Enter price: 50
Enter quantity: 1

Product 2
Enter product name: Perfume
Enter price: 290
Enter quantity: 1

----------------------------------
Total Amount = 340.00
Final Amount = 340.00
----------------------------------
Thank You! Visit Again
```

11. Result

The program executed successfully and produced the expected output, displaying total and final amounts accurately.

12. Applications

- Useful for small grocery or retail stores.
- Learning and understanding the basics of billing systems.
- Academic project for programming practice.

13. Conclusion

The project successfully demonstrates the use of C programming concepts such as input/output, loops, and conditional statements. It provides a simple yet effective way to automate grocery billing.

14. Future Enhancements

- Add file handling for saving bills.
- Improve user interface with better formatting.
- Add options for tax calculation and multiple customer billing.

15. References

- C Programming textbooks.
- Online C programming tutorials.
- Class notes and examples from Programming in C course.